

Software Requirements Specification (SRS)

Secure File Encryption and Decryption System

1. Introduction

1.1 Purpose

The purpose of this document is to define the requirements for a Secure File Encryption and Decryption System. The system allows users to securely encrypt files to protect sensitive data and decrypt them when authorized.

1.2 Scope

This system is designed to provide data confidentiality using cryptographic techniques. Users can select any file, apply encryption using a secret key, and later decrypt the file using the same key. The system is intended for:

- Students
- Individuals
- Small organizations

1.3 Definitions

Encryption: Converting readable data into unreadable format.

Decryption: Converting encrypted data back to original form.

Key: A secret value used for encryption and decryption.

2. Overall Description

2.1 Product Perspective

The system is a standalone desktop application that performs encryption and decryption on local files.

2.2 User Characteristics

Designed for basic computer users with no technical expertise required.

2.3 Operating Environment

Operating System: Windows / Linux / macOS

Programming Language: Python / Java

Libraries: Cryptography library

3. Functional Requirements

FR1 – The system shall allow users to select a file.

FR2 – The system shall allow users to enter a secret key.

FR3 – The system shall encrypt the selected file.

FR4 – The system shall decrypt the encrypted file.

FR5 – The system shall display success or error messages.

FR6 – The system shall prevent decryption with wrong key.

4. Non-Functional Requirements

Security: The system shall not store the secret key and shall use AES-256 encryption.

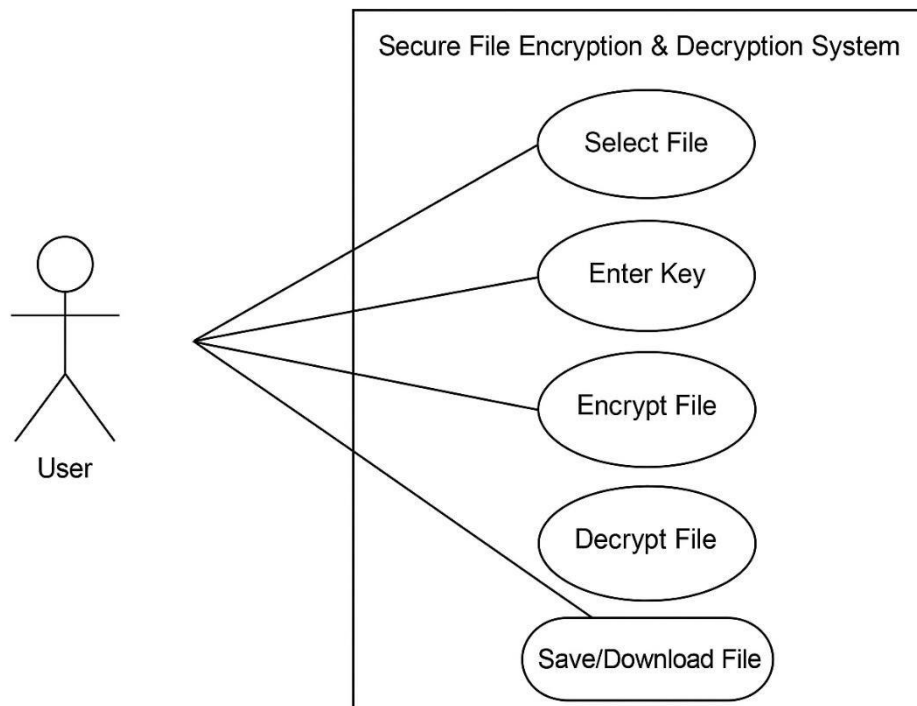
Performance: Encryption should complete within reasonable time and handle files up to 100MB.

Usability: Simple user interface with easy file selection and key input.

5. Use Cases

Encrypt File: User selects file, enters key, system encrypts and saves file.

Decrypt File: User selects encrypted file, enters key, system decrypts and restores original file.



6. Security Considerations

Possible threats include unauthorized access, key guessing, and file tampering.

Security controls include AES-256 encryption, password-based key, no key storage, and input validation.

7. Future Enhancements

- Multi-factor authentication
- Cloud storage encryption
- File integrity verification

